Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/27/2019
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Project Title:	Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	04/15/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	JPL 1571699		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, PK-4, is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments.
The intellectual merit of this research project is centered on three major research topics.

Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination.

Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma.

Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves.

This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discoveries made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental findings that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project makes use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is build by the European Space Agency (ESA), highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involve the development of universal software, hardware, or analysis tools that can also be used under different research situations, and some of the more specific technical tools that are developed, such as a plasma impedance probe, might even help to overall improve technology that can be used for industrial purposes. Finally, as part of the general research mission, we train the next generation of engineers and scientists who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond.

In 2018 the project went into its second year with the workforce at Auburn Univeristy and Wittenberg University fully established. Still the workforce at Wittenberg University grew during the reporting period with the hiring of a second undergraduate student, while at Auburn University we could win a postdoc to partially work on this project starting towards the end of 2018. As in the year before we made wide use of our flourishing cooperation with multiple other institutions. The main cooperation was with the leading PK-4 core team members of the International Space Station (ISS) Experiment "Plasma Kristall 4" (PK-4) localized at the German Aerospace Center (DLR e.V.), Germany. As a result of this cooperation we were able to conduct pre-flight experiments at DLR, to plan and test PK-4 space experiments scenarios that would utilize the PK-4 science reference module. Further, a new cooperation with the dusty plasma research group located at the US partner institution of Baylor University, Texas, was started. Latter cooperation allowed us to coordinate major proposed US experiments for PK-4 to be scheduled under the umbrella of a common experiment scenario. This effort simplifies the organization and planning of US PK-4 experiment contributions. We expect that our planned experiments will be executed in the upcoming PK-4 experiments scheduled for 2019.
The overall work performed under this project in year two was conducted following the outlined plan in the underlying proposal. Some adaptation on the work schedule had to be initiated as the PK-4 space experiment was not operational for most parts of 2017 as well as the beginning of 2018. Due to the effort of DLR/ESA with an experiment upgrade in 2018, PK-4 has become fully operational again mid 2018. Due to the delay in PK-4 experiment conduction we continued in year two of the project with our initial development of tools as seem necessary for the future analysis of the planned PK-4 space experiment. Further we reworked / improved our simulation tools that represent the base of our analysis

Task Progress:

bench-marking system. We tested our ideas, instead of utilizing PK-4 aboard the ISS, by conducting experiments with the ground based PK-4 science reference module, a twin setup of the space experiment located at DLR. We prepared the experiment scenarios by analyzing PK-4 data of experiments that were conducted earlier. The German Aerospace Center kindly supported us with data of latter experiments. We also developed alternative approaches to study the scientific topics of interest and performed related experiments in the lab.

In the following we give a summary of the activities/progress details as archived in year two categorized in three major sub topics:

Topic T1: Charging and interaction of micro-particles in a plasma, Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas, Topic T3: Particle transport and heating in waves,

We have been also involved in activities that go beyond the utilization of the PK-4 space- and ground-based experiments. Breadboard experimental setups, developed for a PK-4 successor experiment “Ekoplasma,” were also utilized. Most of these additional activities have a clear relation to the major research topics T1-T3 and are thus described under the most fitting category. Activities purely related to the development of next generation experiments or those that are related to PK-4 administrative work of the PIs are summarized under:

Topic TX: Future space experiments, administrative tasks, and analysis tool developments.

We were able to archive the following described progress towards our project goals:

Topic T1: A major goal of this project is to investigate the charging of the dust particles and their mutual interaction in the plasma. There are multiple ways that can be followed to derive an appropriate experimentally estimate for the particle charge; for example, by investigating dust-dust particle collisions, studying normal/wave modes in the system of many particles or, via an indirect approach, by determining the discrepancy of electron to ion density that is balanced by the dust particle density times the charge number occupied by each particle. Latter approach was followed by one of our graduate research assistant (GRA). For this approach, we assume global charge neutrality, the sum of all charges to be 0, since otherwise substantial electric fields would arise and work on the system to reestablish charge neutrality. The average dust particle charge is then the missing quantity in the overall charge balance. Measuring the plasma ion density can be performed using Langmuir probes by measuring the ion saturation current of a negatively biased probe. The electron density can in principle be worked out from the current to a Langmuir probe using a positive probe bias. Latter, however, is disturbing the plasma and can lead to contamination of the probe. Instead, we utilize a less invasive technique by measuring the plasma response on a low power electrical stimulus also refereed to as active plasma resonance spectroscopy (APRS). The resonance in the plasma response to the stimulus is related to electron plasma frequency and thus to the electron density.

During year two of the project, the GRA improved this measurement technique and started to automatize it. This will allow us to obtain significantly more data and perform parameter scans in a final stage as planned for year three of this project. In 2018 we also have built a theoretical base for the plasma equivalent circuit model, also involving an undergraduate student. Second, we explored different plasma generation methods from dc to rf (capacitatively-coupled) discharges and their influence on our developed measurement technique. We found that rf driven plasma show disturbing signals in our measurements at frequencies that are multiples of the plasma excitation frequency. The GRA successfully investigated software techniques to clear the data from these disturbances. We also followed an advanced approach by utilizing a single vector network analyzer (VNA) instead of the combination of a spectrum analyzer (SA) and tracking generator (TG). As this measurement is a differential measurement the disturbing signals of the plasma generation naturally disappear. Still, our developed discharge probe represents a disturbance to the plasma in general. We will therefore investigate in the future, if the discharge electrode can be used instead of a Langmuir style discharge probe for this measurements.

A second way that we investigate to determine the particle charge is based on the analysis of the trajectories of dust particles in a dust cloud during close approach interactions (collisions). This technique is especially intended to be applied to the PK-4 data that show rather excited motion of particles where individual particles occasionally have very close approaches. We plan to apply this particle charge measurement technique mainly as a tangential analysis, by searching (data mining) for particle collisions in other, in the first place unrelated, experiments. To establish the analysis we first need to derive criteria that determine under which circumstance a close approach collision can be used to derive a suitable charge estimate. To be able to do so, we need data for which the particle charge is already known in the first place. The analysis on the particle motion has to be applied to this artificial data, a charge value determined, followed by a comparison with the know charge value. To generate the bench-marking data set we have to rely on dusty plasma simulations as the charge on the dust particles are not precisely known in experiments so far. A GRA started in year one setting up a molecular dynamic simulation program, simulating the dynamics of dust particles with a user define charging state. In year two the GRA improved the initial coding of the MD simulation and generated first data sets that show close approach situations that can be used now to study and develop analysis techniques to reconstruct the particle charge from the dynamics and ultimately develop the criteria for this data mining project for PK-4. While continuing in year three with generating base material for the dust-dust collision analysis, the GRA will also study a new, highly related approach studying magnetic field effects on a dust particle flow to derive a charge value for the dust grain from observed density variation in the flow. The recorded particle motion in this experiment together with the derived dust particle charge may be used as an additional benchmark tool for the dust collision analysis. Topics T2/T3: Topics T2 and T3 are performed at a reduced scale compared to the initial proposal. As such we report on the effort on these two topics combined. Both Co-Investigators (CoIs) work as the leading team on these research topics supported full-time since early 2018 and before part time by a GRA (financed from other sources) as well as two (one started in 2018) undergraduate students. We also made substantial use of collaborative support by the PK-4 core group at DLR, a dusty plasma group at the University of Giessen, Germany, and a research group at Baylor University, TX, which work on similar topics utilizing PK-4 and with whom we started a cooperation recently. As s result of our cooperation we could make extensive use of the DLR’s PK-4 science reference module to perform experiments especially related to topics T2/T3 on ground while there was no access to the PK-4 space experiment due to a glitch in its performance.

We spent much of 2018-2019 learning how to prepare experiments for and analyze measurements of the PK-4 setup. For the analysis we used particle image velocimetry (PIV) analysis techniques as well as particle tracking velocimetry (PTV) methods. Using sample data that we acquired from the Science Reference Module in Dec. 2017, we developed several analytical techniques for extracting temporal and spatial information from the flow profiles that suggest a redistribution of flown kinetic energy to thermal energy at the onset of polarity switching. We also compared the results of PTV and PIV methods and identified the constrains for a successful analysis -- “when to use what.”

Working with an undergraduate student from Baylor University in Summer 2018, we further did a complete re-write of our molecular dynamics simulation code related to these experiments. The new code uses a more modern software architecture, is faster, and is natively three-dimensional -- features that enhance our ability to perform better simulations. Using the new MD code and new synthetic image generation, we discovered that the mapping techniques that had previously been developed for extracting velocity distribution information from stationary dusty plasma had to be modified to be used for flowing dusty plasmas, like those in PK-4. This has been a critical element that is necessary to properly interpret PK-4 data.

Topic TX: In addition to the activities above, we also performed some activities towards the development, definition, and utilization of the next generation space experiment for complex plasma research, "EkoPlasma," formerly named "PlasmaLab." Some of our design suggestions were included in the newest Ekoplasma breadboard setup. It is expected that this experiment will be starting its activities aboard the ISS sometimes in the near future. We also made use of the Ekoplasma breadboard setup located at DLR by performing experiment that have been not possible so far in the dusty plasma community. In year one of the project we used the new experimental device to study the behavior of a dusty plasma under an active compression (electrodes are moved during the experiment to reduce the dust cloud volume and thus increase the dust density). In last year's report we showed that as a result of the compression, and thus increase of dust density, the system undergoes a phase transition. The system crystallizes. We investigated our obtained data in year two of the project in more detail and also found that the dynamical behavior has also been changing. We found for example, that the wave lengths of dust lattice waves decrease with increasing dust cloud density. The observations are fascinating and demonstrate the huge potential of the new experiment facility.

Overall: During the second project year we presented our findings and ideas on multiple occasions. Some of the major events where we presented our work: The 15th Workshop on the Physics of Dusty Plasmas, a tri-annular international workshop that represents the second most important event in the dusty plasma field; The 60th Annual Meeting of the American Physical Society, Division of Plasma Physics; the Annual NASA Fundamental Physics Workshops in 2018/2019; and The PK-4 Meetings at the German Aerospace Center. Overall we gave more than 20 presentations as talks or posters during the reporting period and the Principal Investigator (PI)/Co-Is also contributed to some of those events as part of the program committees. Two of the PIs are also members of the PK-4 facility science team (FST) that oversee the scientific mission of PK-4. In summary, we believe that the second year of our project was very successful, which is clearly indicated by the large number of presentations we gave and the extensive communications we had with other researchers. The mentioned glitch in the space experiment setup that was mentioned before and that caused some of our ISS based experiments to be delayed, was resolved in 2018. On the other hand, it also has opened up opportunities for closer collaboration with the German Aerospace Center as well as partner institutions in the US (Baylor) in the attempt to overcome this situation. As a final result, we established a much closer, very flexible cooperation that will positively influence the work to come for all partners.

An outlook: For the third year we plan to perform dedicated experiments aboard the ISS utilizing PK-4 after its recent update. We expect further to obtain additional experimental PK-4 data that can, in an effort of a "tangential analysis," be used to derive scientific results towards our project goals. A comprehensive experiment scenario in cooperation with Baylor University was proposed recently to DLR covering all topics of our proposed research as well as those studied by the group at Baylor University. The suggested experiment schenario is intended to utilize one of the next PK-4 experiments campaigns in 2019. Topics T2,T3 are fully covered with the proposed experiment scenarios in great detail. The analysis related to the dust particle charging T1 will be performed as a tangential analysis on the obtained results. Performing a dedicated experiment related to particle charging, such as individual triggered particle collisions or particle beam dust cloud collisions, is highly complex and thus difficult to realize with the ISS experiment PK-4 for now. We keep this approach as an option for future experiments, and concentrate instead on the tangential analysis of situations where a substantial disorder appears – which is the preferable environment for close approach dust-dust collisions. We expect to report our results in 2019 during the 61st APS-DPP meeting and presumably on some other occasions. The next large conference in the dusty plasma field will be in 2020, the 9th International Conference on Dusty Plasmas (ICPDP) in Moscow.

Besides the activities toward the major goals of the project, also 2019/2020 will contain the PIs' duties as PK-4 facility science team members to oversee the PK-4 mission in general. We will also support DLR as in last year in the definition and development of the next generation complex plasma space experiment "Ekoplasma" and perform, on an occasional basis, experiments with this new setup. On the long term this new experiment, that now is actively pushed by DLR and its partners in Russia for realization, has the potential to transform the dusty plasma research under microgravity condition. It opens the field to large, controllable systems. We are looking forward to an eventful third project year.

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